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Adaptive Tensegrity-Based Control for Multi-Agent Obstacle Avoidance

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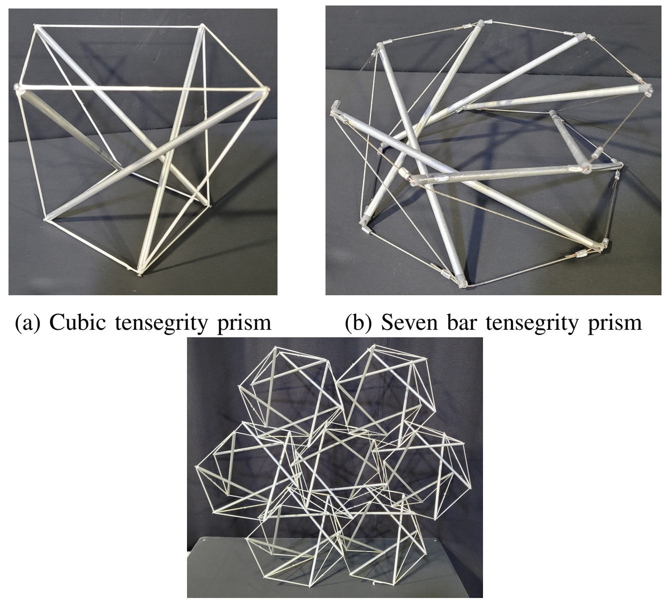
***Abstract*— This paper addresses the problem of multi-agent coordination and obstacle avoidance for systems with limited sensing. Inspired by tensegrity structures, a nonlinear, distance-based control law composed of networked virtual tensile and compressive members is proposed. An adaptive gain relaxation on tensile forces is added, allowing formations to smoothly deform around convex obstacles before reforming back to the designed configuration. The obstacle avoidance protocol only relies on local spatial measurements, making it suitable for agents with limited sensing. Simulations demonstrate that a simple formation utilizing tensegrity-based control successfully navigates obstacles and is robust to noise and dynamic disturbances.**

I. Introduction

The principles of structural rigidity have found growing application in the control of multi-agent systems (MAS). A core challenge of MAS is maintenance of geometric relations between agents, especially in formations with limited sensing and communication. When each agent only has access to local measurements – being the relative position of neighboring agents and nearby obstacles – formation control strategies must preserve structure and adaptability without relying on global information or centralized control. By viewing these networks as a graph configuration, rigidity theory can be applied to control strategies to preserve the formation integrity in the presence of disturbances. This is beneficial for MAS networks as it increases redundancy, robustness, and configuration flexibility [1], [2], [3], [4]. However, while many practical MAS applications require a fixed configuration, complex environments often demand adaptability. Obstacles, constraints, and dynamic mission objectives necessitate formations to deform or reconfigure while maintaining their overall rigidity properties [5]. Integrating deformation capabilities for obstacle and collision avoidance into the MAS control law is essential to guarantee safety and mission success.



(a) Cubic tensegrity prism

(b) Seven bar tensegrity prism

(c) Six bar tiled tensegrity prism plate

Fig. 1: Tensegrity structures use tension to stabilize compression members. This arrangement offers potential for high-precision control in robotics and automated systems.

In this paper, we consider a class of structural frameworks called *tensegrity structures*, which are systems of purely compressive and tensile members held together in equilibrium [6]. In mechanical systems, these members are realized as bars (compressive) and strings (tensile), as seen in Figure 1. Based on these principles, we produce virtual tensegrity structures by defining each agent as a node of a structure and imposing attractive (tensile) and repulsive (compressive) forces between agents. These structures then define a control law that maintains geometric relations between agents while being resilient to noise and disturbances.

An effective MAS formation must be able to navigate through environments with obstacles. Looking to nature we see large systems of independent agents such as birds or fish that are able to morph and split around hazards and obstacles, and many MAS control algorithms have taken inspiration from these [7], [8]. However, these algorithms do not maintain formation topology, so a different formulation is used.

The musculoskeletal system of many vertebrates and tensegrity structures have often been related and studied

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together [6], [9]. On the micro level, the cytoskeletal system of single cell organisms can be modeled as a tensegrity structure [10]. Examining flexibility in these systems, the skeletal system of many snakes have ligaments that can relax during feeding, allowing it to eat prey larger than its head, such as the common watersnake. As a result, the ligament connecting the two halves of the snakes' mandible can separate by 770% its resting length [11]. On the micro level, mutable collagenous tissue (MCT), found in the skin sea cucumbers and other members of the Echinodermata phylum can be rigid via tension between collagen fibrils, protecting against predators. By chemically relaxing the stress between fibrils, this tissue becomes soft allowing for locomotion [12], [13], [14].

Inspired by these examples, virtual string relaxation is introduced in the proposed MAS control algorithm, allowing the formation to loosen when deformed past equilibrium. This provides the formation with the ability to navigate through obstructed environments while maintaining formation topology.

Related Work

The balance between rigidity and deformation control makes a tensegrity-based framework an attractive candidate for designing robust and adaptive MAS control frameworks. Several approaches have been proposed to implement tensegrity-based formation control, e.g. [15], [16], [17], [18], [19]. These methodologies model formations with virtual forces acting like pre-stressed linear springs that are functions of inter-agent distances. While the proposed formulations capture the essential behavior of tensegrity elements, the inclusion of pre-stressed springs can lead to negative axial stiffness in the virtual force domain, thereby producing bistability and potential non-unique equilibrium configurations. Furthermore, the restriction to linear springs limits the controller design. In [20], a nonlinear, power law-based forcing function is utilized. However, this method relies on a set rest lengths of each member, which removes the natural pre-stress property of the tensegrity framework and creates a fragile equilibrium set. Research specializing in collision and obstacle avoidance in tensegrity-based formation control is relatively scarce. A linear, string-based tensegrity framework that can vary the lengths of the virtual bars to avoid obstacles is proposed in [19]. However, this work relies on pre-planned length adjustments from a mission planner and does not consider collision avoidance sensing or an adaptive obstacle avoidance mechanism.

Main Contributions

In this paper, a generalized tensegrity-based formation control algorithm is proposed for MAS with limited sensing in an obstructed environment. Compared with the results discussed in the literature review, the contributions of this paper are threefold. First, a distributed formation control law is formulated using power-law action functions. These virtual forces do not rely on preset member lengths, and their linearity is guided by the function parameters, increasing freedom in controller performance design. Second, a collision avoidance protocol is integrated to ensure that agents do not collide with each other or obstacles. Lastly, a tension relaxation behavior is introduced. A continuous piecewise function is utilized to ease virtual string tensions in the presence of increased inter-agent distances. This smoothly adjusts the stiffness of the formation, allowing the agents to spread out due to navigation and collision avoidance forces while preserving the tensegrity-based graph topology. Numerical simulations validate these theoretical findings and demonstrate convergence of the MAS formation to a desired equilibrium configuration.

This manuscript is organized as follows. Section II discusses preliminary concepts of graph theory and tensegrity principles, as well as notation. The problem statement is outlined in Section III. Our main result is presented in Section IV, which consists of the novel forcing functions used in the proposed tensegrity-based control law, as well as the obstacle avoidance and string relaxation functions. Section V constructs tensegrity based formations and provides simulations of the response to navigating an environment with obstacles. Finally, Section VI provides concluding remarks and future directions.

II. Preliminaries and Problem Statement

A. Graph Theory

Let n denote the number of nodes embedded in an d-dimensional configuration. Let $\{q_i \in \mathbb{R}^d\}_{i\in\mathcal{V}}$ denote the position of a node and define the node matrix $q = [q_1, q_2, ..., q_n] \in \mathbb{R}^{d\times n}$. An *undirected graph* $\mathcal{G}$ is a pair $(\mathcal{V}, \mathcal{E})$ that consists of a set of nodes $\mathcal{V} = \{1, ..., n\}$ and a set of edges $\mathcal{E} \subseteq \mathcal{V}\times\mathcal{V}$ where information flows equally in both directions across each edge. The set of neighbors of node i is defined by $\mathcal{N}_i = \{j \in \mathcal{V} : a_{ij} \neq 0\} = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$.

The *adjacency matrix* $A = [a_{ij}]$ of an undirected graph is a symmetric matrix with nonzero elements satisfying the property $a_{ij} \neq 0 \Leftrightarrow (i, j) \in \mathcal{E}$ and otherwise $a_{ij} = 0$ and $a_{ii} = 0$. This matrix is used for describing the communication weights among nodes. The *degree matrix* $D = D(A)$ is a diagonal matrix with elements $D_{ii} = \sum_{j=1}^{n} a_{ij}$. The *graph Laplacian* L is defined as

$$L = D(A) - A \tag{1}$$

The matrix L always has a uniform eigenvector $\mathbf{1}_n = (1, ..., 1)^\top$ associated with a $\lambda = 0$ eigenvalue. The undirected graph $\mathcal{G}$ is considered *connected* iff $\text{rank}(L) = n-1$.

B. Tensegrity Structures and Self-Equilibrium

The necessary condition for a structure to be stable is that its stiffness matrix K with respect to any non-trivial motion $\boldsymbol{v}$, satisfies the equation

$$\boldsymbol{v}^\top K\boldsymbol{v} > 0. \tag{2}$$

The matrix K of a tensegrity structure can be written as

$$K = K_E + K_G \tag{3}$$

where K_E is the linear stiffness matrix and K_G is the geometrical stiffness matrix. Assume that the tensile forces in the edges have positive axial stiffness, such that K_E is always positive semi-definite. Then, Equation (2) holds only if K_G is also positive semi-definite. This is determined by the sufficient conditions for structural stability, which are dependent on two quantities: the force density matrix and the geometry matrix, which are described next.

Let f_{ij} be the internal force of the edge connecting the i-th and j-th nodes. Further, let $\ell_{ij} = ||q_i - q_j||$ be the length of that edge. Denote $\omega_{ij} = f_{ij}/\ell_{ij}$ as a *tension coefficient*. Then, the *force density matrix* (FDM), $\mathcal{D} \in \mathbb{R}^{n\times n}$, is a weighted Laplacian defined as

$$[\mathcal{D}]_{ij} = \begin{cases} -\omega_{ij}, & \text{if } i \neq j \\ \sum_{k\in\mathcal{N}_i} \omega_{ik}, & \text{if } i = j \\ 0, & \text{if } i \notin \mathcal{N}_j \end{cases} \tag{4}$$

The symmetric matrix $\mathcal{D}$ represents the weighted connectivity of the structure and is used to evaluate its prestress stability. Form-finding analysis uses this matrix to find self-equilibrium (self-stress) configurations. The nodal coordinates of the equilibrium configuration can be found using the set of equations

$$\mathcal{D}\boldsymbol{x} = 0, \qquad \mathcal{D}\boldsymbol{y} = 0, \qquad \mathcal{D}\boldsymbol{z} = 0 \tag{5}$$

where $\boldsymbol{x}$, $\boldsymbol{y}$, and $\boldsymbol{z}$ are the solution coordinate vectors.

For a structure with $\mathcal{B}$ bars and $\mathcal{S}$ strings, let $C_B \in \mathbb{R}^{\mathcal{B}\times n}$ and $C_S \in \mathbb{R}^{\mathcal{S}\times n}$ denote the associated connectivity matrices:

$$[C_B]_{ij}, [C_S]_{ij} = \begin{cases} -1, & \text{if } q_i \text{ is the start node of member } j \\ 1, & \text{if } q_i \text{ is the end node of member } j \\ 0, & \text{if } q_i \text{ is not connected to member } j \end{cases}$$

Define a combined connectivity matrix $C = \begin{bmatrix} C_B^\top & C_S^\top \end{bmatrix}^\top$. Then, the coordinate differences of members in each dimension may be written as

$$\boldsymbol{u} = C\boldsymbol{x}, \qquad \boldsymbol{v} = C\boldsymbol{y}, \qquad \boldsymbol{w} = C\boldsymbol{z} \tag{6}$$

Denote the diagonal form of $\boldsymbol{u}$ by $\boldsymbol{U} = \text{diag}(\boldsymbol{u}) \in \mathbb{R}^{(\mathcal{B}+\mathcal{S})\times(\mathcal{B}+\mathcal{S})}$ and similarly for $\boldsymbol{V} = \text{diag}(\boldsymbol{v})$ and $\boldsymbol{W} = \text{diag}(\boldsymbol{w})$. The *geometry matrix*, G, is defined as

$$G = \begin{bmatrix} \boldsymbol{U}\boldsymbol{u} & \boldsymbol{V}\boldsymbol{v} & \boldsymbol{W}\boldsymbol{w} & \boldsymbol{U}\boldsymbol{v} & \boldsymbol{U}\boldsymbol{w} & \boldsymbol{V}\boldsymbol{w} \end{bmatrix} \tag{7}$$

Then, it is possible to define sufficient conditions that must be met for a stable tensegrity structure.

Lemma 1: A tensegrity structure embedded in d dimensions is stable if the following conditions are satisfied:

1) *The FDM $\mathcal{D}$ has a rank deficiency of at least $d + 1$.*
2) *$\mathcal{D}$ is positive semi-definite.*
3) *The rank of the geometry matrix G is $d(d + 1)/2$.*

Proof: The proof follows from [21]. ■

III. Problem Formulation

Consider each agent in a homogeneous MAS as a distinct node with each inter-agent interaction described as an edge in the undirected graph $\mathcal{G}$. Assume that agent $i \in \mathcal{V} = \{1, ..., n\}$ can measure the distance between itself and a neighbor, given by $q_i - q_j$, $j \in \mathcal{N}_i$. Furthermore, the agent can measure obstacles within the set radius r_y, found within the obstacle set $\mathcal{N}_i^c$. Let $\boldsymbol{d}_j$ denote the position of the closest point of an obstacle to the agent, with $j \in \mathcal{N}_i^c$. The dynamics of agent i are described as

$$u_i = u_i^t(q) + \sum_{j\in\mathcal{N}_i^c} u_{ij}^c(q_i, \boldsymbol{d}_j, r_y) + u_i^n \tag{8}$$

$$m\ddot{q}_i = -c\dot{q}_i + \text{sat}_{u_{\max}}(u_i) \tag{9}$$

where m is the mass of the agent, $c > 0$ is a damping parameter, u_i^t is the edge force term, u_{ij}^c is the collision avoidance term, and u_i^n is the navigation term. The latter input represents an external input that steers the agent according to a higher-level navigational control law. Finally, the magnitude of the resultant summation is clamped to $\pm u_{\max}$ which represents input saturation.

The objective is to design the distributed control law u_i^t such that the formation converges to a desired equilibrium configuration while resisting small perturbations and maintaining the geometric structure, as shown in Figure 2. The desired outcome is then

$$q_i(t) \to q_i^*, \quad \text{for } i = 1, \ldots, n$$

where $q^* = [q_1^*, \ldots, q_n^*]$ is a nominal self-stress state consistent with the tensegrity structure. Due to the limited sensing capabilities of the collision avoidance term, each agent can only obtain local obstacle information. Consequently, trajectory planning-based obstacle avoidance is not feasible. The control objective is thus extended to allow the equilibrium condition of the formation to expand, allowing the system to adaptively deform and move around obstacles, while preserving formation geometry.

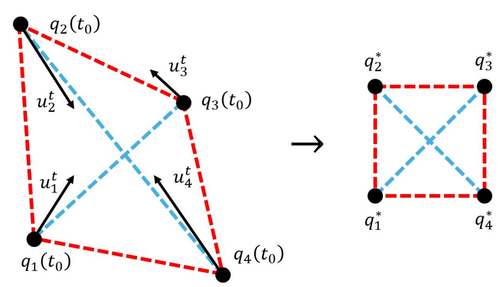


Fig. 2: Coordination and geometry of a MAS formation. Blue and red lines represent bar and string virtual forces, respectively. The objective is to design the distributed control law such that the formation converges to a desired equilibrium configuration.

IV. Main Results

This section presents the novel tensegrity-based MAS control law. First, Section IV-A presents the design of the virtual tensegrity structure force model that establishes a state of self-stress in the MAS. Next, Section IV-B introduces a string relaxation protocol for obstacle avoidance.

A. Tensegrity-Based Formation Control

Let $\mathcal{E}_B \subset \mathcal{E}$ denote the set of bar edges, and define the bar neighbor set of node i as

$$\mathcal{N}_i^B := \{\, j \in \mathcal{N} \mid (i,j) \in \mathcal{E}_B \}$$

Similarly, let $\mathcal{E}_S \subset \mathcal{E}$ denote the set of string edges, with neighbor set

$$\mathcal{N}_i^S := \{\, j \in \mathcal{N} \mid (i,j) \in \mathcal{E}_S \}$$

Define an internal edge force as

$$f_{ij}(z) = k_{ij} z^{\alpha_{ij}} \tag{10}$$

where $k_{ij} \in \mathbb{R}$ and $\alpha_{ij} \in \mathbb{R}$. Denote k_B and α_B as bar parameters and k_S and α_S as string parameters such that

$$\{k_{ij}, \alpha_{ij}\} = \begin{cases} \{k_B, \alpha_B\}, & \text{if } (i,j) \in \mathcal{E}_B, \\ \{k_S, \alpha_S\}, & \text{if } (i,j) \in \mathcal{E}_S, \\ \{0, 0\}, & \text{if } i = j \end{cases} \tag{11}$$

Define the collective potential function

$$V(q) = \sum_{(i,j)\in\mathcal{E}} \psi(||q_i - q_j||) \tag{12}$$

where

$$\psi(z) = \int_0^z f(s) ds \tag{13}$$

Inputs to a node from virtual bar and string connections can be derived by taking the gradient of the potential function. The edge force term takes the form

$$\begin{aligned} u_i^t &= -\nabla_{q_i} V(q) \\ &= -\sum_{j\in\mathcal{N}_i} f(\|q_i - q_j\|)\boldsymbol{n}_{ij} = -\sum_{j\in\mathcal{N}_i} k_{ij}\ell_{ij}^{\alpha_{ij}}\boldsymbol{n}_{ij} \end{aligned} \tag{14}$$

where

$$\ell_{ij} = ||q_i - q_j|| \quad \text{ and } \quad \boldsymbol{n}_{ij} = \frac{q_i - q_j}{\ell_{ij}}$$

Using Lemma 1, stability conditions of this system with respect to the bar and string parameters may be derived.

Lemma 2: Consider a system of agents in a configuration that admits a state of self-stress, q^*, and a positive semi-definite FDM $\mathcal{D}$. Assume that $u_{ij}^c = 0 \quad \forall \{i,j\}$ and $u_i^n = 0 \ \forall i$. Then the formation is structurally stable under the condition

$$k_{ij}(\alpha_{ij} - 1) \geq 0 \qquad \forall (i,j) \in \mathcal{E}$$

or equivalently:

1) *If $\alpha_{ij} > 1$, then $k_{ij} \geq 0$.*
2) *If $\alpha_{ij} = 1$, then $k_{ij} \in \mathbb{R}$.*
3) *If $\alpha_{ij} < 1$, then $k_{ij} \leq 0$*

Proof: The stiffness matrix K is defined as the Hessian of the system evaluated at an equilibrium point, $K = \nabla^2 V(q^*) \in \mathbb{R}^{dn\times dn}$. Consider a block Hessian matrix $H_{ij} \in \mathbb{R}^{d\times d}$ such that K is in the form

$$K = \begin{bmatrix} H_{11} & H_{12} & \cdots & H_{1n} \\ H_{21} & H_{22} & \cdots & H_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ H_{n1} & H_{n2} & \cdots & H_{nn} \end{bmatrix} \tag{15}$$

Each block is found to be

$$H_{ij} = \begin{cases} -\omega_{ij}(I_d + (\alpha_{ij} - 1)\boldsymbol{n}_{ij}^*\boldsymbol{n}_{ij}^{*\top}), & \text{if } i \neq j \\ \sum_{k\in\mathcal{N}_i} \omega_{ik}(I_d + (\alpha_{ik} - 1)\boldsymbol{n}_{ik}^*\boldsymbol{n}_{ik}^{*\top}), & \text{if } i = j \end{cases}$$

where $I_d \in \mathbb{R}^{d\times d}$ is the identity matrix and $\boldsymbol{n}_{ij}^* = \boldsymbol{n}_{ij}(q^*)$. Equation (3) is now recovered with $K_G = \mathcal{D} \otimes I_d$, with $\otimes$ denoting the Kronecker product, and

$$K_E = \begin{cases} -\omega_{ij}(\alpha_{ij} - 1)\boldsymbol{n}_{ij}^*\boldsymbol{n}_{ij}^{*\top}, & \text{if } i \neq j \\ \sum_{k\in\mathcal{N}_i} \omega_{ik}(\alpha_{ik} - 1)\boldsymbol{n}_{ik}^*\boldsymbol{n}_{ik}^{*\top}, & \text{if } i = j \end{cases} \tag{16}$$

Since it is given that $\mathcal{D}$ is positive semi-definite, then K_G is also positive semi-definite. Therefore, $K_E \succeq 0 \Leftrightarrow K \succeq 0$. Let $v \in \mathbb{R}^{dn} \neq 0$ be an arbitrary vector where

$$v^\top K_E v = \sum_{i\neq j} \omega_{ij}\ell_{ij}^{-2}(\alpha_{ij} - 1)\left((q_i^* - q_j^*)^\top (v_i - v_j)\right)^2$$

As a result, K_E is positive semi-definite under the condition that $\omega_{ij}\ell_{ij}^{-2}(\alpha_{ij} - 1) = k_{ij}\ell_{ij}^{\alpha_{ij}-3}(\alpha_{ij} - 1) \geq 0$. Since $\ell_{ij} \geq 0$, the condition reduces to $k_{ij}(\alpha_{ij} - 1) \geq 0$. ■

Remark 1. For the tensegrity-based formation, it is assumed that the connectivity matrices are chosen such that a state of self-stress exists. Under this condition, the internal force densities induced by the bar and string elements yield a positive, semi-definite FDM. Hence, the positive semi-definiteness of $\mathcal{D}$ follows directly from the chosen connectivity and the assignment of tensile and compressive members.

Remark 2. This assignment further constrains the gains k_B and k_S to ensure the correct internal tension or compression along bars and strings. Thus, from Equation (10), bar edges must enforce $k_B < 0$ and string edges must enforce $k_S > 0$. Consequently, to preserve formation stability according to the conditions found in Lemma 2, $\alpha_B < 0$ and $\alpha_S \geq 1$.

B. Expansion for Obstacle Avoidance

In this subsection, an extension of the tensegrity-based MAS control scheme is presented. The approach enables agents to avoid collisions and adjust the formation's equilibrium condition to move around larger obstacles. Let the unit vector between agent i and obstacle j be defined as

$$\boldsymbol{p}_{ij} = \frac{q_i - \boldsymbol{d}_j}{||q_i - \boldsymbol{d}_j||} \tag{17}$$

When obstacle j is another agent, then $\boldsymbol{p}_{ij} = \boldsymbol{n}_{ij}$. Define the collision avoidance term

$$u_{ij}^c = -k_a \left(||q_i - \boldsymbol{d}_j||^{-\gamma} - r_y^{-\gamma}\right) \boldsymbol{p}_{ij} \tag{18}$$

where $k_a > 0$ and $\gamma > 0$ are constants.

An additional challenge arises when navigating around obstacles, as the formation must be capable of splitting and subsequently reforming. To enable this behavior, Equation (10) is reformulated as a piecewise function that relaxes the virtual string forces once the formation has progressed beyond a specified boundary point as

if $\{i, j\} \in \mathcal{E}_S$

$$f_{ij}(\ell_{ij}) = \begin{cases} k_{ij}\ell_{ij}^{\alpha_{ij}} & \ell_{ij} \leq z_1 \\ h(\ell_{ij}) & z_1 \leq \ell_{ij} \leq z_2 \\ \beta & z \geq z_2 \end{cases} \tag{19}$$

where $h(\ell_{ij})$ is a cubic spline smoothly connecting the piecewise function defined as

$$\begin{aligned} h(\ell_{ij}) = (2s^3 - 3s^2 + 1)k_{ij}z_1^{\alpha_{ij}} \\ + (s^3 - 2s^2 + s)(z_2 - z_1)k_{ij}z_1^{\alpha_{ij}-1}\alpha_{ij} \\ + (-2s^3 + 3s^2)\,\beta \end{aligned} \tag{20}$$

with $s = \frac{\ell_{ij} - z_1}{z_2 - z_1}$ and β is a constant.

As seen in Figure 3, the terms z_1 and z_2 are the constant dividers of the piecewise function where the function has a constant value, β, after z_2.

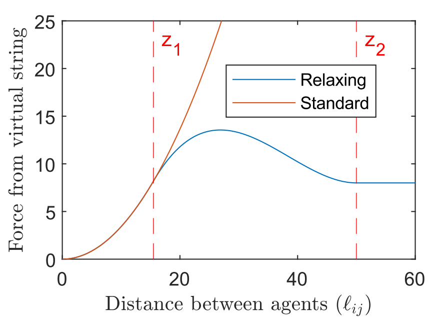


Fig. 3: String force vs. inter-agent distance for standard and relaxing string

V. Numerical Results

In this section, the performance of the tensegrity-based control law and the obstacle avoidance protocol is demonstrated. Consider a $d = 2$ dimensional square-shaped network of four agents with the connectivity of bars and strings, as shown in Figure 2. A tiled formation made up of six interconnected squares (two rows and three columns) is used in the simulation. For this particular formation, the equilibrium condition can be found analytically. Denote l_S as the length of the string connecting two agents along the perimeter of a single square tile. For this formation, a force balance yields the following ratio:

$$\frac{k_B}{k_S} = -l_S^{\alpha_S - \alpha_B}\sqrt{2}^{-\alpha_B + 1} \tag{21}$$

By choosing l_S, α_S, and α_B, the gains may be found. For this study, these values were chosen heuristically through simulation.

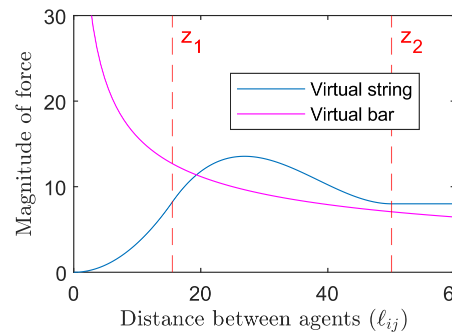
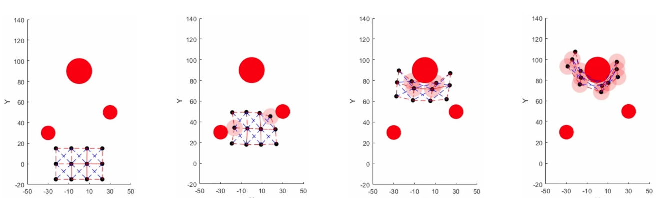


Fig. 4: Magnitude of force from virtual Bar and String

The values used for each parameter in the simulation are found in Table I. The parameter z_1 was chosen to be slightly greater than l_S so the formation does not start relaxing when exposed to small deformations. The parameters z_2 and b were chosen such that the virtual string force magnitude is always larger than the virtual bar force magnitude when $\ell_{ij} >> l_S$, as shown in Figure 4. This is to keep the formation from diverging when the strings relax. In addition, Gaussian noise with a standard deviation of $\sigma = 0.5$ was imposed on all measurements.

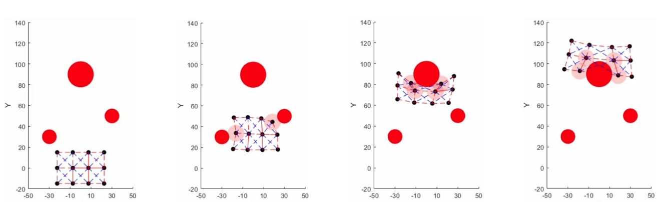
TABLE I: Constants used in simulations

dt	0.05	k_S	0.0341
m	1	k_B	-50
u_i^n	[0,3]	α_S	2
c	1.5	α_B	-0.5
l_S	15	z_1	15.5
σ	0.5	z_2	50
r_y	8	β	8
k_a	20	γ	0.4
u_{max}	10		



(a) $t = 0$ s (b) $t = 15$ s (c) $t = 30$ s (d) $t = 40$ s

Fig. 5: Simulation of tensegrity formation without string relaxation



(a) $t = 0$ s (b) $t = 15$ s (c) $t = 30$ s (d) $t = 40$ s

Fig. 6: Simulation of tensegrity formation with string relaxation

The tiled formation consisting of 12 agents is shown in Figure 5 for the scenario when the strings do not relax and in Figure 6 for the scenario when string relaxation is adopted. Both formations use tensegrity-based control; a comparison to a linear spring-based control method is found in [22]. Solid red circles denote obstacles in the environment. When a translucent red circle appears over an agent, it has detected an object within its avoidance radius r_y and is reacting to it. A constant navigational force of $u_i^n = [0, 3]$ is applied to all agents of the formation, forcing the formation to move in the positive Y direction.

As designed, both formations behave similarly while navigating close to obstacles demanding small deformations. However, as shown in Figure 5, when the strings do not relax, the formation can get hung up while navigating around large obstacles. On the other hand, the formation enhanced with string relaxation, shown in Figure 6, can navigate around the large obstacle by having a large local deformation of the two center units while keeping the integrity of the edge units. This behavior is achieved without communication of the obstacle presence between agents. A video comparing these two simulations can be found at
`https://youtu.be/25G0i-iGIP4`.

VI. CONCLUSIONS

This paper presented a tensegrity-based formation control framework designed for MASs with limited sensing and communication capabilities. By combining nonlinear power-law interactions with adaptive string relaxation, the proposed approach enables cohesive formations that can smoothly deform to navigate convex obstacles and reform once the obstacles are clear, without requiring global information or centralized coordination. The control law relies only on local distance and obstacle measurements, making it lightweight and suitable for implementation on agents with restricted computational and sensing resources. Simulations demonstrate robustness to noise and dynamic disturbances while preserving overall formation integrity. Tensegrity-based formation control, with continued development, is a viable and versatile control method that fuses the structural properties of tensegrity structures into MAS control.

A. Current Research Directions

These algorithms are currently being implemented onto hardware in order to study the formation's behavior under measurement and process noise. Mathematically proving the stability and uniqueness of tensegrity formations is also ongoing research. With these results, 3-D formations and their behavior will be studied more fully. In addition, self organization of large scale formations will be studied. Agents will be modeled as rigid bodies rather than point masses in order to help generalize the problem. We will apply this algorithm to space-like, forced environments rather than the flat plane studied in this paper.

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